

Eigenvalues, Eigenvectors & Inner Product Spaces

A rigorous, intuition-first treatment of two foundational pillars of linear algebra — with worked examples, geometric interpretation, and direct connections to machine learning algorithms.

Part A — Slides 1–6

Eigenvector intuition · The equation $Av = \lambda v$ · Diagonal matrix example · Characteristic equation · Eigenvector extraction

Part B — Slides 7–12

Inner product spaces · Dot product geometry · Orthogonality · Projection · ML applications: neural networks, cosine similarity, embeddings, RAG · Concept map summary

Rigorous

All math verified correct with full worked derivations

Geometric

Every concept grounded in visual, coordinate-plane intuition

Applied

Direct links to PCA, neural nets, embeddings, and Transformers

What Happens When a Matrix Transforms a Vector?

A matrix **A** acts as a *transformation machine* — it takes an input vector and produces an output vector, generally in a completely different direction. Understanding what these transformations do is the starting point for eigenvector theory.

Four Things a Matrix Can Do

Stretch

Makes the vector longer — scale factor > 1

Compress

Makes the vector shorter — $0 < \text{scale} < 1$

Rotate

Changes the direction of the vector entirely

Reflect

Flips the vector across an axis

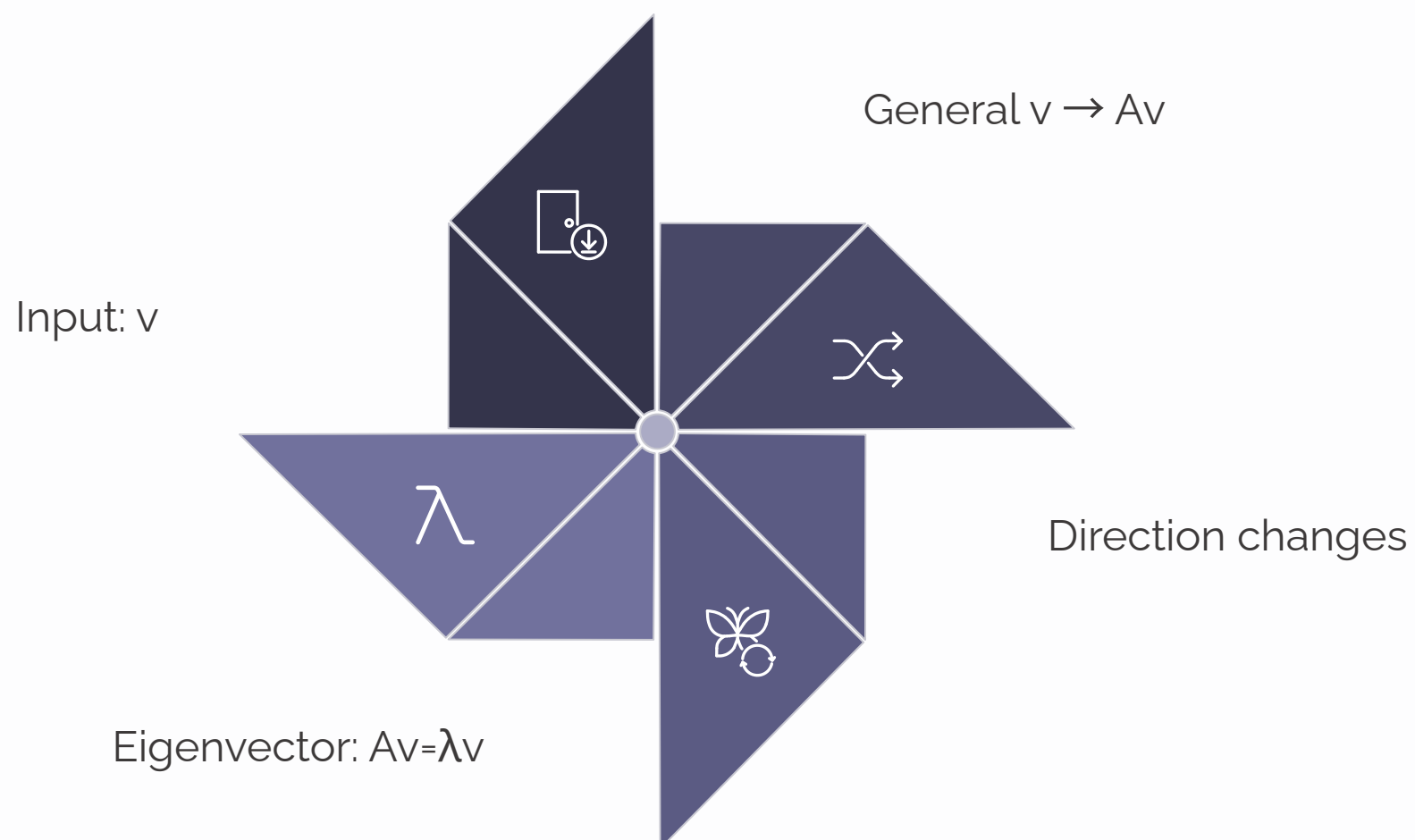
The Key Question

For *most* vectors, multiplication by **A** sends them in a completely new direction. The transformed output has almost nothing in common directionally with the input.

? Is there any special vector whose direction does NOT change after multiplication by **A?**

- Yes — and these are called **Eigenvectors**. Their direction is preserved under the transformation; only their length (magnitude) may change. This is the central question of eigenanalysis.

The word "eigen" comes from German, meaning *own* or *characteristic* — an eigenvector is, in a sense, the matrix's own natural direction.



Eigenvector Intuition: $Av = \lambda v$

The Fundamental Equation

$Av = \lambda v$

A — the transformation matrix

An $n \times n$ matrix representing a linear map from $\mathbb{R}^n$ to $\mathbb{R}^n$

v — the eigenvector

A special, non-zero vector whose direction survives the transformation

λ (**lambda**) — the eigenvalue

A scalar (just a number) that tells us the scaling factor applied along this direction

Plain English Intuition

- An **eigenvector** is a special direction that *survives* the transformation — it does not rotate, only scales
- The **eigenvalue** tells us *how much* that direction is stretched ($\lambda > 1$), compressed ($0 < \lambda < 1$), flipped ($\lambda < 0$), or unchanged ($\lambda = 1$)
- Think of it as: the matrix "agrees" with this vector's direction and simply scales it — a perfect resonance between matrix and vector

Visual Flow

Input	Operation	Output
v	Multiply by A	$Av = \lambda v$
Direction: preserved	Length: scaled by λ	Same line through origin

$$\lambda=1$$

Unchanged

Vector stays identical — no stretch, no flip

$$\lambda>1$$

Stretched

Longer but same direction — amplified

$$\lambda<0$$

Flipped

Direction reversed — pointing the other way

$$\lambda=0$$

Collapsed

Vector maps to zero — matrix is singular

Worked Example: Diagonal Matrix

Diagonal matrices are the simplest and most instructive case. Their eigenvectors are the coordinate axes themselves, and their eigenvalues are simply the diagonal entries.

The Matrix

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$$

Eigenvector 1 | $v_1 = [1, 0]^T$

$$Av_1 = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \cdot [1, 0]^T = [3, 0]^T = 3 \cdot [1, 0]^T$$

- ✓ Direction unchanged — only stretched by 3
- $\lambda_1 = 3$ | $v_1 = [1, 0]^T$ (horizontal axis)

Eigenvector 2 | $v_2 = [0, 1]^T$

$$Av_2 = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix} \cdot [0, 1]^T = [0, 2]^T = 2 \cdot [0, 1]^T$$

- ✓ Direction unchanged — only stretched by 2
- $\lambda_2 = 2$ | $v_2 = [0, 1]^T$ (vertical axis)

Step-by-Step Verification

01

Write the diagonal matrix A

$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}$ — off-diagonal entries are zero

02

Multiply A by the candidate eigenvector

$A \cdot [1, 0]^T$: first row gives $3 \cdot 1 + 0 \cdot 0 = 3$; second row gives $0 \cdot 1 + 2 \cdot 0 = 0 \rightarrow [3, 0]^T$

03

Check that output = $\lambda \cdot$ (same vector)

$[3, 0]^T = 3 \cdot [1, 0]^T$ ✓ — confirms $\lambda_1 = 3, v_1 = [1, 0]^T$

04

Repeat for the second eigenvector

$A \cdot [0, 1]^T = [0, 2]^T = 2 \cdot [0, 1]^T$ ✓ — confirms $\lambda_2 = 2, v_2 = [0, 1]^T$

📌 **Interpretation:** This diagonal matrix simply stretches the x-axis by 3 and the y-axis by 2. The coordinate axes are the natural eigenvectors — a unit square transforms into a 3×2 rectangle.

How to Find Eigenvalues: The Characteristic Equation

The Key Formula

$$\det(\mathbf{A} - \lambda \mathbf{I}) = 0$$

$\mathbf{A}$ = the matrix

The square matrix whose eigensystem we want

λ = unknown eigenvalue

The scalar we are solving for

$\mathbf{I}$ = identity matrix

We subtract λ from each diagonal entry

$\det = 0$ condition

This forces a polynomial — solving it gives all eigenvalues

For an $n \times n$ matrix, the characteristic polynomial has degree n — an $n \times n$ matrix has exactly n eigenvalues (counted with multiplicity, over $\mathbb{C}$).

Worked Example | $\mathbf{A} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

01

Form $\mathbf{A} - \lambda \mathbf{I}$

$$\begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

02

Compute determinant

$$(4-\lambda)(3-\lambda) - (1)(2) = 0$$

03

Expand the product

$$12 - 4\lambda - 3\lambda + \lambda^2 - 2 = 0 \rightarrow \lambda^2 - 7\lambda + 10 = 0$$

04

Factor the polynomial

$$(\lambda - 5)(\lambda - 2) = 0$$

05

✓ Read off eigenvalues

$$\lambda_1 = 5 \mid \lambda_2 = 2$$

📄 **Interpretation:** This matrix has two special scaling directions — one stretches by a factor of 5, the other by a factor of 2. We now use these eigenvalues to find the actual eigenvectors (next slide).

How to Find Eigenvectors: $(A - \lambda I)v = 0$

Once eigenvalues are known, substitute each λ back into the system and solve the *homogeneous* linear system $(A - \lambda I)v = 0$. The non-trivial solutions are the eigenvectors.

For $\lambda_1 = 5$ | $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

- $A - 5I = \begin{bmatrix} 4-5 & 1 \\ 2 & 3-5 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$
- Row 1: $-x + y = 0 \rightarrow y = x$
- Row 2: $2x - 2y = 0 \rightarrow$ same relation (rows are dependent, as expected)
- Let $x = 1$: $\checkmark \mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}^T$

This vector points at 45° — the matrix stretches it by a factor of 5.

Verification

$$A \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}^T = \begin{bmatrix} 4+1 \\ 2+3 \end{bmatrix}^T = \begin{bmatrix} 5 \\ 5 \end{bmatrix}^T = 5 \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix}^T \quad \checkmark$$

For $\lambda_2 = 2$ | $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

- $A - 2I = \begin{bmatrix} 4-2 & 1 \\ 2 & 3-2 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$
- Row 1: $2x + y = 0 \rightarrow y = -2x$
- Row 2: identical — one free variable, as expected
- Let $x = 1$: $\checkmark \mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}^T$

This vector points down-right — the matrix stretches it by a factor of 2.

Verification

$$A \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix}^T = \begin{bmatrix} 4-2 \\ 2-6 \end{bmatrix}^T = \begin{bmatrix} 2 \\ -4 \end{bmatrix}^T = 2 \cdot \begin{bmatrix} 1 \\ -2 \end{bmatrix}^T \quad \checkmark$$

Key Method Insight

The rows of $(A - \lambda I)$ are always linearly dependent when λ is a true eigenvalue — that's why $\det = 0$ and why a non-trivial solution exists

Eigenvectors Are Not Unique

Any scalar multiple of an eigenvector is also an eigenvector. We typically report the simplest representative (e.g., $x = 1$)

Eigenspace

The set of all eigenvectors for a given λ , plus the zero vector, forms a subspace called the eigenspace of λ

From Vectors to Inner Products

The inner product generalises the notion of "how much two vectors agree" — it simultaneously encodes length, angle, and orthogonality. It is the single most-used operation in machine learning.

Definition — Inner Product (Dot Product)

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^T \mathbf{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n$$

What It Gives Us

Similarity

How much do two vectors "agree" in direction? Large positive = strong alignment

Angle

The angle θ between two vectors is encoded: $\mathbf{u}^T \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$

Length (Norm)

$\|\mathbf{v}\| = \sqrt{\mathbf{v}^T \mathbf{v}}$ — the length of a vector is its inner product with itself

Orthogonality

If $\mathbf{u}^T \mathbf{v} = 0$, the vectors are perpendicular — completely independent directions

Why It Matters in Machine Learning



Neural Networks

Every neuron computes a dot product: $\mathbf{z} = \mathbf{w}^T \mathbf{x} + \mathbf{b}$. The inner product measures how well the input aligns with the learned weight vector before the activation function decides whether the neuron fires.



Embeddings & Similarity

Words, sentences, images, and documents are vectors in high-dimensional space. Similarity between any two embeddings is measured using their inner product — the foundation of semantic search and recommendation.



Transformers / Attention

Attention scores in self-attention are dot products between query and key vectors: $\mathbf{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathbf{softmax}(\mathbf{QK}^T / \sqrt{d}) \mathbf{V}$. Inner products are the computational engine of modern LLMs.

Dot Product & Geometric Meaning

The Formula

$$\mathbf{u}^T \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

The dot product equals the product of the vector lengths times the cosine of the angle between them. This is the bridge between algebra (numbers) and geometry (angles and directions).

Three Critical Cases

Sign of $\mathbf{u}^T \mathbf{v}$	Angle θ	Interpretation
● $\mathbf{u}^T \mathbf{v} > 0$	$\theta < 90^\circ$	Similar direction — positive alignment
● $\mathbf{u}^T \mathbf{v} = 0$	$\theta = 90^\circ$	Perpendicular — orthogonal, no shared component
● $\mathbf{u}^T \mathbf{v} < 0$	$\theta > 90^\circ$	Opposite directions — negative alignment

Key Result: Orthogonality Condition

$$\mathbf{u}^T \mathbf{v} = 0 \iff \mathbf{u} \perp \mathbf{v}$$

Orthogonal vectors carry completely independent information — a concept central to PCA, QR decomposition, and Gram-Schmidt orthogonalisation.

Numerical Example

Let $\mathbf{u} = [1, 0]^T$ and $\mathbf{v} = [0, 1]^T$

- $\mathbf{u}^T \mathbf{v} = (1)(0) + (0)(1) = 0 \rightarrow$ orthogonal ✓
- $\|\mathbf{u}\| = 1, \|\mathbf{v}\| = 1 \rightarrow$ both unit vectors
- $\cos \theta = 0 / (1 \times 1) = 0 \rightarrow \theta = 90^\circ$ ✓

Cosine Similarity — The Normalised Version

$$\cos \theta = \mathbf{u}^T \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$$

By dividing by the norms, we remove magnitude bias and measure *pure directional alignment* — always between -1 and $+1$. This is the definition of cosine similarity used universally in NLP and recommendation systems.

Acute Angle

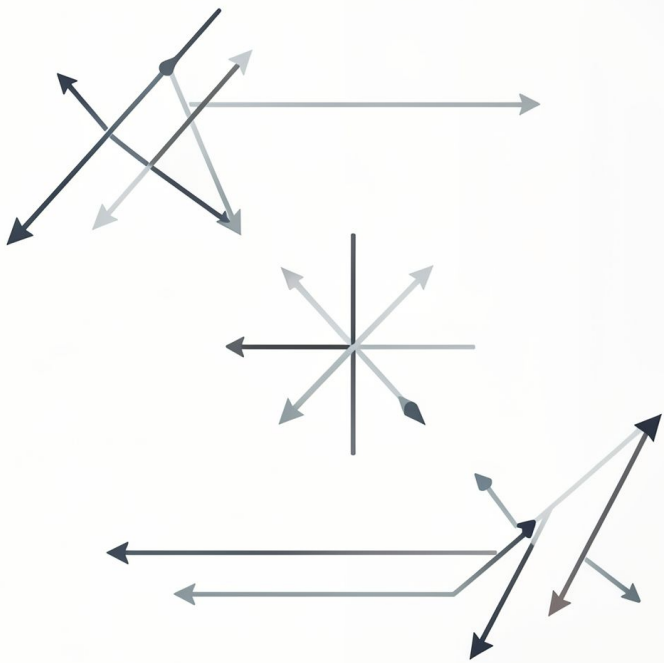
$\mathbf{u} \cdot \mathbf{v}$ greater than 0,
similar direction

Obtuse / Opposite

$\mathbf{u} \cdot \mathbf{v}$ less than 0,
opposite direction

Right Angle

$\mathbf{u} \cdot \mathbf{v} = 0$, orthogonal



Worked Example: Inner Product, Length & Angle

Given Vectors

$\mathbf{u} = [3, 4]^T \quad \mathbf{v} = [4, -3]^T$

Step 1 — Inner Product

$\mathbf{u}^T \mathbf{v} = (3)(4) + (4)(-3) = 12 - 12 = 0$

✔️ **✓ $\mathbf{u} \perp \mathbf{v}$ — the vectors are perpendicular!**

Step 2 — Lengths (Norms)

$\|\mathbf{u}\| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

$\|\mathbf{v}\| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$

- Both vectors have equal length = 5
- They form a perfect right angle in $\mathbb{R}^2$

Interpretation

$\mathbf{u}$ and $\mathbf{v}$ are equal-length, perpendicular vectors. They form a perfect right angle. This is confirmed by $\mathbf{u}^T \mathbf{v} = 0$ and visually by the 90° angle on the coordinate plane.

Full Summary Table

Quantity	Formula	Value
Inner product $\mathbf{u}^T \mathbf{v}$	$(3)(4) + (4)(-3)$	0
Norm $\ \mathbf{u}\ $	$\sqrt{9 + 16}$	5
Norm $\ \mathbf{v}\ $	$\sqrt{16 + 9}$	5
Angle θ	$\arccos(0 / 25)$	90°
Orthogonal?	$\mathbf{u}^T \mathbf{v} = 0?$	✓ Yes

Key Takeaway

The inner product is zero precisely when the angle is 90° . You don't need to compute the angle explicitly — checking $\mathbf{u}^T \mathbf{v} = 0$ is both necessary and sufficient for orthogonality. This algebraic shortcut is used constantly in ML.

0

Inner product $\mathbf{u}^T \mathbf{v}$

Confirms perfect orthogonality — no shared directional component

5

$\|\mathbf{u}\| = \|\mathbf{v}\|$

Both vectors have identical Euclidean length — equal magnitude

90°

Angle θ

A right angle between $\mathbf{u}$ and $\mathbf{v}$ — completely independent directions

Projection & Orthogonal Decomposition

Projection answers: *how much of vector u lies along the direction of v ?* — like a shadow cast by u onto the line defined by v . The result is a vector in the direction of v .

The Projection Formula

$$\text{proj}_v(u) = (u^T v / v^T v) \cdot v$$

Worked Example | $u = [3, 4]^T$, $v = [1, 1]^T$

01

Compute $u^T v$

$$(3)(1) + (4)(1) = 7$$

02

Compute $v^T v$

$$(1)(1) + (1)(1) = 2$$

03

Compute scalar factor

$$u^T v / v^T v = 7/2 = 3.5$$

04

Multiply by v

$$\text{proj}_v(u) = 3.5 \cdot [1, 1]^T = \checkmark [3.5, 3.5]^T$$

Orthogonal Decomposition

Every vector u can be split into two orthogonal components:

Component	Vector	Meaning
Along v	$[3.5, 3.5]^T$	The projection — what u "contributes" in direction v
Perpendicular to v	$[-0.5, 0.5]^T$	$u - \text{proj}_v(u)$ — the residual / error component
Sum	$[3, 4]^T = u$	The two components reconstruct u exactly $\checkmark$

Verification of Orthogonality

$$[3.5, 3.5]^T \cdot [-0.5, 0.5]^T = (3.5)(-0.5) + (3.5)(0.5) = -1.75 + 1.75 = 0 \quad \checkmark$$

- ☐ **ML Applications:** Projection is the core idea behind Gram-Schmidt orthogonalisation, QR decomposition, least-squares regression (finding the closest point in a subspace), and PCA (projecting data onto principal eigenvector directions).

Inner Products in Machine Learning

1 — Neural Networks: Every Neuron Computes a Dot Product

$$\mathbf{z} = \mathbf{w}^T \mathbf{x} + \mathbf{b}$$

- $\mathbf{w}$ = weight vector (learned parameters)
- $\mathbf{x}$ = input feature vector
- $\mathbf{b}$ = bias scalar
- $\mathbf{w}^T \mathbf{x}$ is a dot product — it measures how well the input aligns with the learned weights
- The activation function $\sigma(\mathbf{z})$ then decides whether the neuron "fires"

2 — Embeddings & Cosine Similarity

Words, sentences, images, and documents are represented as high-dimensional vectors (embeddings). Similarity is measured by:

$$\cos \theta = \mathbf{u}^T \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$$

cos θ value	Interpretation
$\approx +1$	Very similar — nearly identical direction
≈ 0	Unrelated — orthogonal in embedding space
≈ -1	Opposite meaning — antonyms / contradictions

Four Major Application Domains



Semantic Search

Find documents closest to a query in embedding space. The query vector and document vectors are compared by cosine similarity — the top-k results are returned.



Recommendation Systems

User and item embeddings are compared by inner product. Items whose vectors align most closely with the user's preference vector are recommended.



RAG — Retrieval-Augmented Generation

Relevant document chunks are retrieved from a vector store via cosine similarity before being passed to an LLM as context. Inner products are the retrieval mechanism.



Transformers / Attention

Attention scores are dot products between query $\mathbf{Q}$ and key $\mathbf{K}$ vectors, scaled by $\sqrt{d}$: $\text{softmax}(\mathbf{Q}\mathbf{K}^T/\sqrt{d})\mathbf{V}$. Every attention head in GPT-4 runs millions of dot products per token.

Big Picture: Key Takeaways

◆ Eigenvalues & Eigenvectors

$$Av = \lambda v$$

Special directions preserved under a matrix transformation — only scaled, never rotated

Eigenvalue λ

The scaling factor — how much that direction is stretched, compressed, or flipped

$$\det(A - \lambda I) = 0$$

The characteristic equation — yields all eigenvalues as roots of a polynomial

$$(A - \lambda I)v = 0$$

Solve for eigenvectors by substituting each eigenvalue back into the homogeneous system

◆ Inner Product Spaces

Inner Product $u^T v$

Measures alignment/relationship between two vectors — the engine of ML computation

$$\text{Norm } \|v\| = \sqrt{v^T v}$$

Measures the length of a vector — used in normalisation and cosine similarity

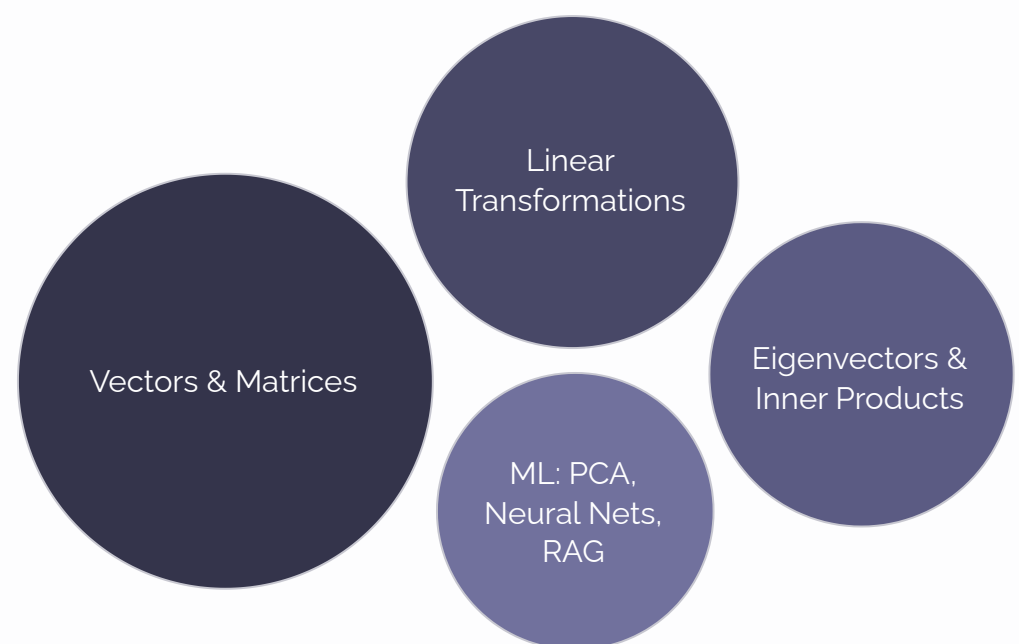
$$\text{Orthogonality: } u^T v = 0$$

Vectors are perpendicular — independent directions, zero shared information

$$\text{Projection: } (u^T v / v^T v) \cdot v$$

Component of u along direction v — basis of least-squares, QR, and PCA

The ML Pipeline



Six Things to Remember

1

Eigenvectors *preserve direction* under a transformation

2

Eigenvalues tell *how much* that direction scales

3

Inner products measure *relationships* between vectors

4

Dot product = 0 means *orthogonality*

5

Projection extracts the *component along a direction*

6

Dot products are the *fundamental operation* in neural networks and embedding-based ML